

SkyGuard

Finding the weather station that is lying, using only the three numbers it is allowed to send.

Everything here is running and measured unless it is marked otherwise. Where a number is weak we print the weak number — a brief that only contains good news tells you nothing about the parts that were actually tested.

01 What was asked, and what exists

The problem	Detect abnormal, inconsistent or faulty observations from Automatic Weather Stations in real time, using only temperature (°C), atmospheric pressure (hPa) and relative humidity (%). Separate genuine weather from sensor faults, minimise false alarms, scale to a national network.
What we built	A running service that grades a 344-station network every step, decides whether to dispatch a technician, and shows its full reasoning for every verdict — deployed, public, and reachable from a browser now.
The central idea	Compare a station with its neighbours at the same moment. Weather is shared between nearby stations and cancels in the difference; a fault belongs to one instrument and does not.
The hard case	Slow calibration drift. No single reading is implausible, so no threshold fires — the failure is the divergence, not the value.
Is it ML?	Not in the decision path today. The intelligence is in multi-agent arbitration and spatial statistics; the learned reference model is designed, marked PLANNED throughout, and drawn dashed in the diagrams.

02 Why thresholds cannot do this

A probe that fails slowly reads a little wrong, then a little more, for weeks. Every reading it produces is a perfectly ordinary number.

That is the case that matters, because it is the one that silently corrupts a climate record and a forecast at the same time. A range check never fires. A rate-of-change check never fires. The station looks healthy right up until somebody compares it with a calibrated reference — which, for most of a national network, nobody does for months.

Divergence is only visible against something that has not diverged. So the reference is the *neighbours*, at the same timestep. If five stations around this one all report an afternoon 0.85 °C warmer than their own normal and this one reports 2.90 °C warmer, the weather cancels and what is left belongs to the instrument.

THE LIMIT, STATED BEFORE ANYONE ASKS

A station with fewer than three usable neighbours cannot be judged this way. We label those NO NEIGHBOURS rather than guessing — visible on the Andaman & Nicobar stations. Admitting where the method cannot reach is what makes the rest of it trustworthy.

03 The method, in five steps

Each step consumes the previous one's output. All of it is arithmetic a judge can check by hand, and the worked example below does exactly that.

1 Remove each station's own climatology, by hour

A mast at 4,196 m is colder than neighbours at 200 m and warms faster at dawn. Differencing raw values flagged **49% of healthy stations**. Subtracting each station's own clean-period hourly mean centres everything near zero.

2 Subtract the median of the neighbours

Median, not mean: the median of five neighbours is unmoved by one of them being broken. What remains is the residual $\bar{x}$ — this station's disagreement with its region.

3 Scale it by that station's own quiet spread

From a robust MAD over a *trailing, lagged* 15-day window, shrunk halfway toward the network median. Trailing and lagged because a drift that enters its own reference window inflates the scale and hides itself.

4 Band at 6σ and 8σ

WATCH and FAULT. Six sounds extreme against the usual 3σ intuition — at 344 stations and 96 steps a day, 3σ alone would produce roughly 90 false flags per day. The bands were calibrated against a fault-free control run, not assumed.

5 Open an episode, with hysteresis

Three consecutive bad steps to raise, six clean steps to clear. An alert is a *duration*, not a spike, which is what makes it a work order rather than a notification.

WORKED – ONE TIMESTEP, TEMPERATURE ANOMALIES IN °C

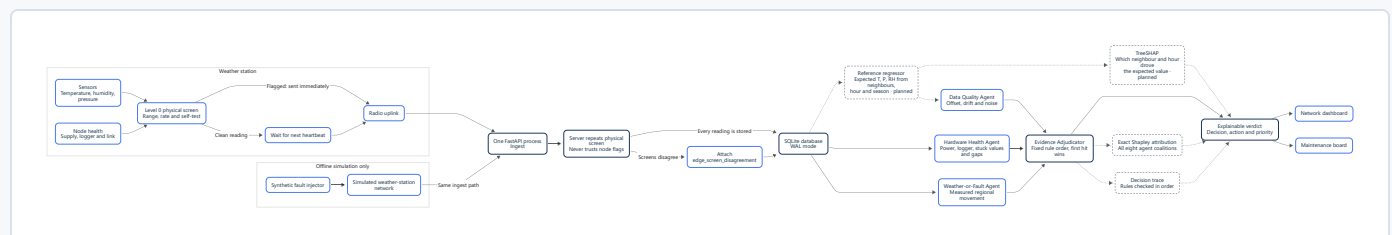
Neighbour anomalies	+0.80 · +0.92 · +0.71 · +1.08 · +0.85
Their median	+0.85
Healthy station reads +0.90, so $r =$	+0.05
Same station with a +2.0 °C offset fault reads +2.90, so $r =$	+2.05
With $sd = 0.3212$ °C, $z = 2.05 / 0.3212$	6.23 → WATCH

Read the healthy row carefully: every station in the region was about 0.85 °C above its own normal — a genuine warm day — and the weather cancelled *completely*, leaving 0.05. The faulty row kept almost the entire 2.0 °C of the fault. That contrast is the whole method.

04 The system, end to end

Sensor to work order. One FastAPI process serves the API, the live socket and the compiled dashboard; there is no separate frontend host and no database server.

WIDE FIGURE – SCROLL SIDEWAYS



Two things are worth pointing at. **The screen sits on the station**, and what it decides is *when to spend radio power*, not what to keep — a suspect reading is sent at once, a clean one waits for the heartbeat, and both reach the server, which repeats the same screen rather than trusting the node. **Flags, never filters.** And the panel is the shared door: the simulated network and the live node are judged by the same three agents and the same adjudicator. Dashed is planned, not running.

05 Who decides, and how it explains itself

Three agents look at non-overlapping evidence. One adjudicator checks a fixed order of rules and stops at the first that fires. Nothing here is an LLM, and nothing trains online.

AGENT	SEES	ANSWERS
Data Quality	The station's residual against its neighbours	Is this an offset, a drift, or noise?
Hardware Health	Supply, logger, link, stuck values, gaps	Is the instrument itself unwell?
Weather or Fault	The neighbours' own movement — independent of the residual	Did the region move, the same way, by enough to explain this?

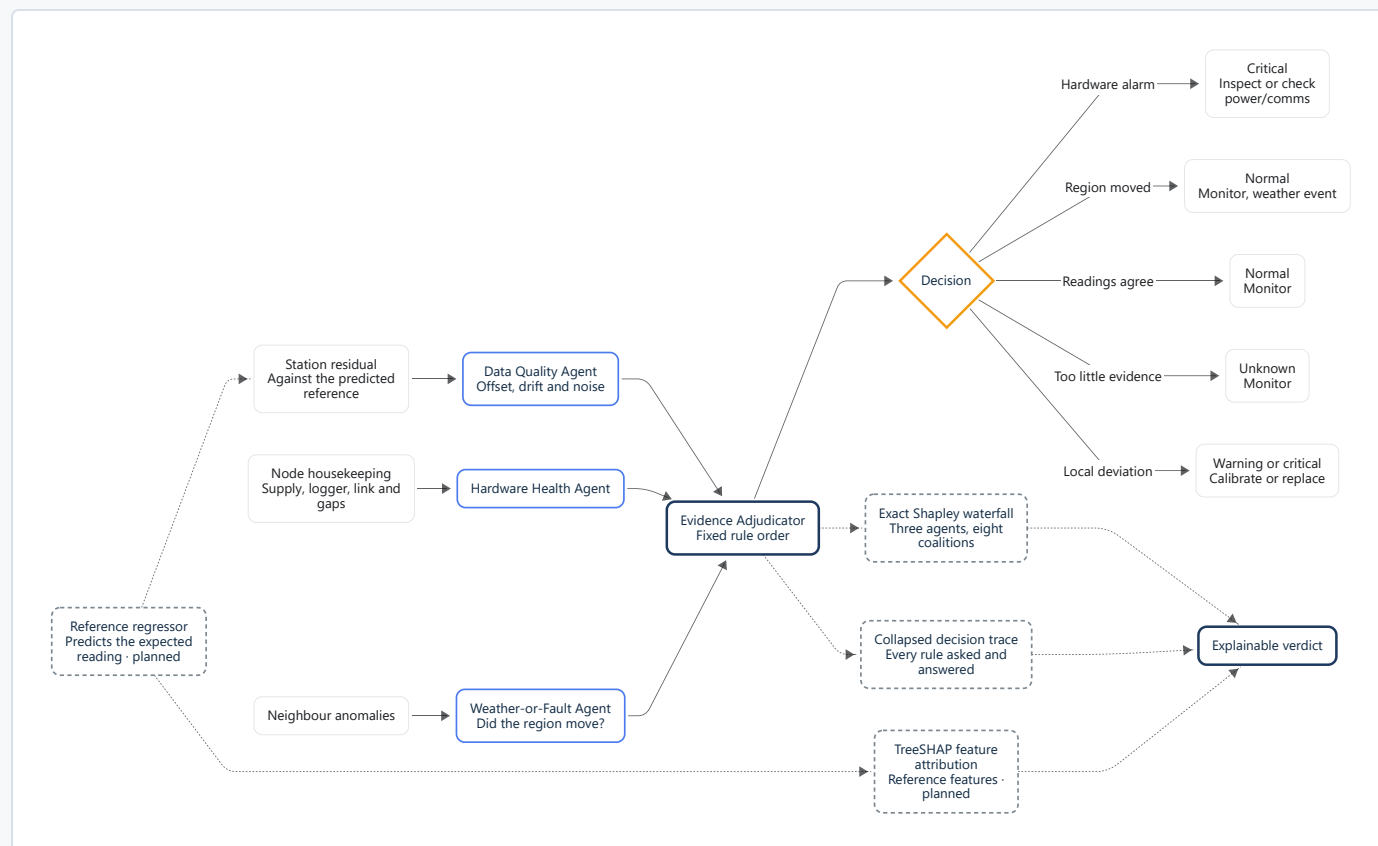
Why exact Shapley, and not SHAP

The problem statement names SHAP as preferable. SHAP is an *approximation* of the Shapley value, and it approximates only because a model has hundreds of features and enumerating 2^n coalitions is impossible. We have three agents. 2^3 is eight. Every coalition is evaluated, so what the board shows is the exact Shapley value from the original theorem — costing three adjudications more than the verdict itself, and summing to the verdict exactly.

WORKED — MANDI, RELATIVE HUMIDITY, FROM THE DEPLOYED SERVICE

Data quality — reading outside the band	$\phi = +0.25$
Hardware health — nothing to look at	$\phi = 0.00$
Weather or fault — 15 neighbours moved 3.7σ	$\phi = -0.25$
Sum = the escalation as issued → MONITOR	0.00

Read the third row. The weather check did not abstain — it argued *against* dispatch by exactly as much as data quality argued for it, and won. A label saying “data quality carried this” cannot express that; a signed number can. Each agent also ships the counterfactual: the action the panel would reach with that agent silenced. “*Silence hardware and this becomes CALIBRATE*” is the half an operator acts on.



The reasoning layer. The dashed boxes are the explanation: every verdict is re-run on all eight coalitions, and the adjudicator records the walk it took, three-valued so a check whose evidence is missing reads “cannot tell” rather than “no”. The reference regressor and the TreeSHAP box beside it are the planned second layer — note that TreeSHAP is fed by the regressor, not by the adjudicator, because the two attribute different quantities.

06 What it actually measures

28 faults injected into a 344-station network across 33 states, 30 days at 15-minute cadence, with the grader given the readings and nothing else. Because we generate the truth, recall and precision are exact.

57% RECALL injected faults found	38% PRECISION of alerting stations	72% RECALL EXCL. DROPOUTS silence has no residual	73 ALERT EPISODES across 42 stations
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Both recall numbers are published because the difference is the honest part: a station that stops reporting produces nothing to difference, so counting dropouts as detection failures blames the detector for a communications outage. The amplitudes were also chosen to run *below* what a 30-day record can support — an easier injector would have bought a better headline and proved less.

The weather veto, measured rather than asserted

The third agent's job is to say “this is weather, do not send anyone.” It earns that by three conditions at once: the region moved at least 2σ , it moved the *same way*, and it moved by at least 40% of this station's excursion. The third clause is the one that matters — a region that moved 2σ while the station moved 30 does not excuse the station.

MEASURED AGAINST THE INJECTED TRUTH — ALL 60 FLAGGED ROWS	
Vetoed issued (“this is weather”)	9
Correct — no fault had been injected	8
Veto precision	89%
False alarms removed	8 of 46 — 17%
Real faults wrongly vetoed	1

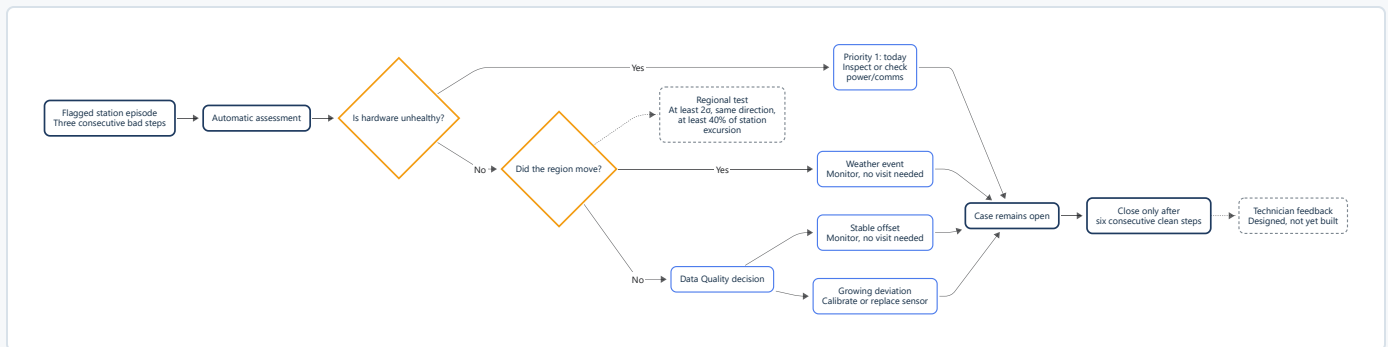
The single miss is a dropout, and the adjudicator alarms on it anyway because hardware is checked *before* the weather veto — a dead link is a fault in any weather. That ordering was

chosen for a different reason and happens to cover this case, which is worth knowing rather than claiming as design.

07 What the operator does with it

The most valuable thing this system does is tell you when *not* to send the van. Every verdict resolves to one of four actions, and three of them cost nothing.

WIDE FIGURE – SCROLL SIDEWAYS



An episode opens on three consecutive bad steps and closes only after six clean ones.

Hardware is asked first because a dead link is a fault in any weather; the regional test is asked next because it is the only question that can cancel a dispatch. A case stays open until the station is quiet again, not until someone dismisses it.

08 The edge tier

An ESP32 sketch runs the same level-0 screen on the station and posts through the same `/api/ingest` door the rest of the network uses. It runs end to end in Wokwi over TLS; it has never run on a physical board, and we say so.

Radio transmission dominates an AWS power budget, so the node's real decision is *when to spend it*. The firmware reads `if (flagged || due)`: a suspect reading is transmitted immediately, a clean one waits for the heartbeat. A flag **brings the uplink forward** — it never suppresses it.

FLAGS, NEVER FILTERS

Nothing is discarded at the node or at the server. The server recomputes the same screen rather than trusting the node — a corrupted screen is itself one of the faults being looked for — attaches `edge_screen_disagreement` where the two disagree, and stores the reading either way. Discard at the edge and nothing downstream can reason about the shape of a fault, because it never received the fault.

The node also keeps a running distribution of its own readings in a P^2 quantile sketch: **20 floats instead of 720 stored samples**, agreeing with the exact quantiles to 99.2–99.4% at the 3σ mark.

09 Built, planned, and not built

Stated plainly so nobody discovers it as a surprise in the demo.

PIECE	STATE	NOTE
Detection pipeline, episodes, dashboard	BUILT	Running and measured; every number in §06 came from the deployed service.
Three agents, adjudicator, exact Shapley, decision trace	BUILT	Visible per station on the maintenance board.
ESP32 node through the real ingest path	BUILT	In Wokwi, over TLS. Never on physical hardware.
Learned reference model + TreeSHAP over it	PLANNED	The place a model belongs is predicting what a station <i>should</i> have read, so the anomaly is the subtraction and TreeSHAP explains the prediction feature by feature. Designed, not trained — there is no <code>shap</code> import in the repository.
An autonomous explanation trajectory	NOT BUILT	The trace walks a <i>fixed</i> tree — two distinct paths across sixty board rows. It is an honest printout of the rules, and it is not an investigation that chooses its next probe.
Faithfulness testing	NOT BUILT	Nothing yet verifies the explanations are not lying. We inject the faults, so “does the top-attributed factor match the injected cause?” is measurable here in a way it is not for most entrants.

ON THE DATA

Station *locations* are real IMD sites; the readings are generated, and stay generated through the finals. That is deliberate: no labelled fault list for Indian AWS data is published, so a generated network is the only way to state recall and precision as facts rather than impressions. What is being demonstrated is the pipeline that judges the readings.

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Where to see it

Live service	Dev-66/skyguard-api — public. Serves the API and the dashboard at <code>/app</code> .
Source	Devansh-66/skyguard — branch <code>main</code> .
Full detail	SkyGuard As-Built for the mathematics with worked examples, and SkyGuard Blueprint for the design reasoning and the traps. Where the two disagree, the as-built is right.
On the board	Open any flagged station: the Shapley waterfall, each agent's counterfactual, and the adjudicator's own walk are on the page, not in a report.